

Compact Laser MBE System

(Model: PAC-LMBE)

System Manual



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1 Introduction

1.1 Outline of the PLD system

PLD system is a thin-film deposition machine which has sintered target sources and a substrate. Fundamental process of the PLD is ablation on a source target surface by exposure of high-energy pulsed laser beam and sticking precursors from the source target to a substrate material. For convenience of thin-film deposition process the PLD system has some units, for example, a source-target manipulator or a substrate heating unit.

Some units in the PLD system have occasions to injure operators or to damage the lab facility when user mistakes. See following chapters and attachment documents which show functions of individual units and parts in detail.

1.2 Caution

	Class 4 laser equipments are used in the PLD system. Keep away from the laser equipments and its beam path during laser emission. Wear protection glasses and clothes if user has to observe inside of the chamber or approach the laser beam path.
	X-ray is generated when an electron beam is irradiated to a substrate material with RHEED (Reflection High Energy Electron Diffraction) gun. Keep away from viewing ports which have no protection.
	Numerous electric power lines with large current or high voltage and sensor signal cables are wired at/to terminals or connectors. Do not touch the lines/cables while the PLD system runs. Shut down the electric power when user has to do cable works.
	As for an optional use of harmful or toxic gases, our original system does not equip any safety protections including a gas detector. On the occasion of use these kinds of gases, a leakage check and accompanying measure against accidents should be done thoroughly by the user.
	This system cannot sustain a normal operation under lack cooling water flow rate. An inspection of the compressed air and cooling water is regularly required.
	This system basically requires being electrified continuously. Connecting the system to a power line of insufficient capacity may cause an accidental electric power failure or system malfunction. The rating of electric power line should be sufficient.
	We do not undertake a maintenance service or a repairing in the condition that the equipment contains harmful, toxic substances or products and specified chemical substances etc. These substances should be completely removed prior to the services.

2 Hardware system guide of the PLD system

PLD system is a thin-film deposition machine which has sintered target sources and a substrate. Fundamental process of the PLD is ablation on a source target surface by exposure of high-energy pulsed laser beam and sticking precursors from the source target to a substrate material. For convenience of thin-film deposition process the PLD system has some units, for example, a source-target manipulator or a substrate heating unit. Overview of the PLD systems is shown in Figure 1.

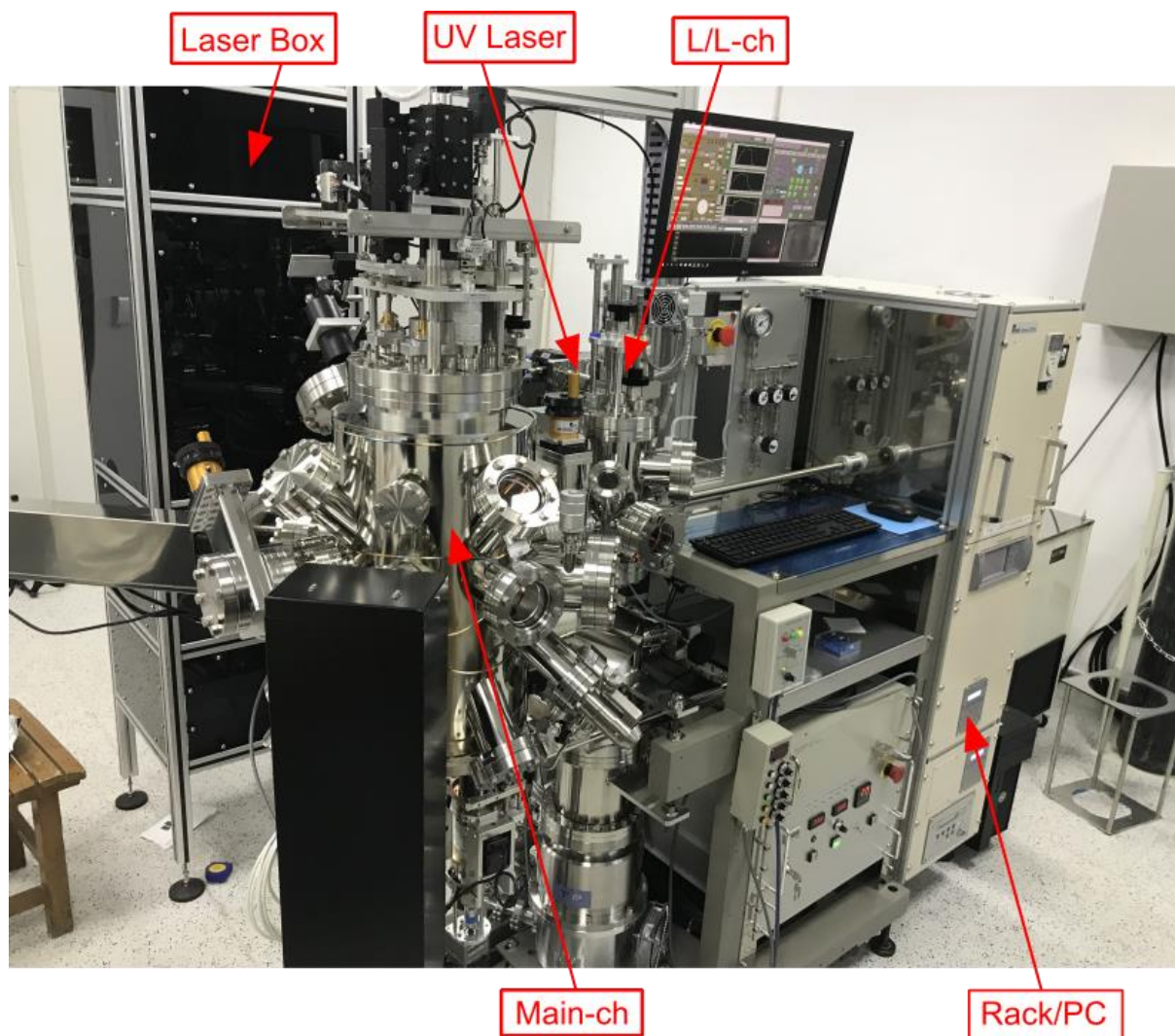


Figure 1: Overview of the PLD system.

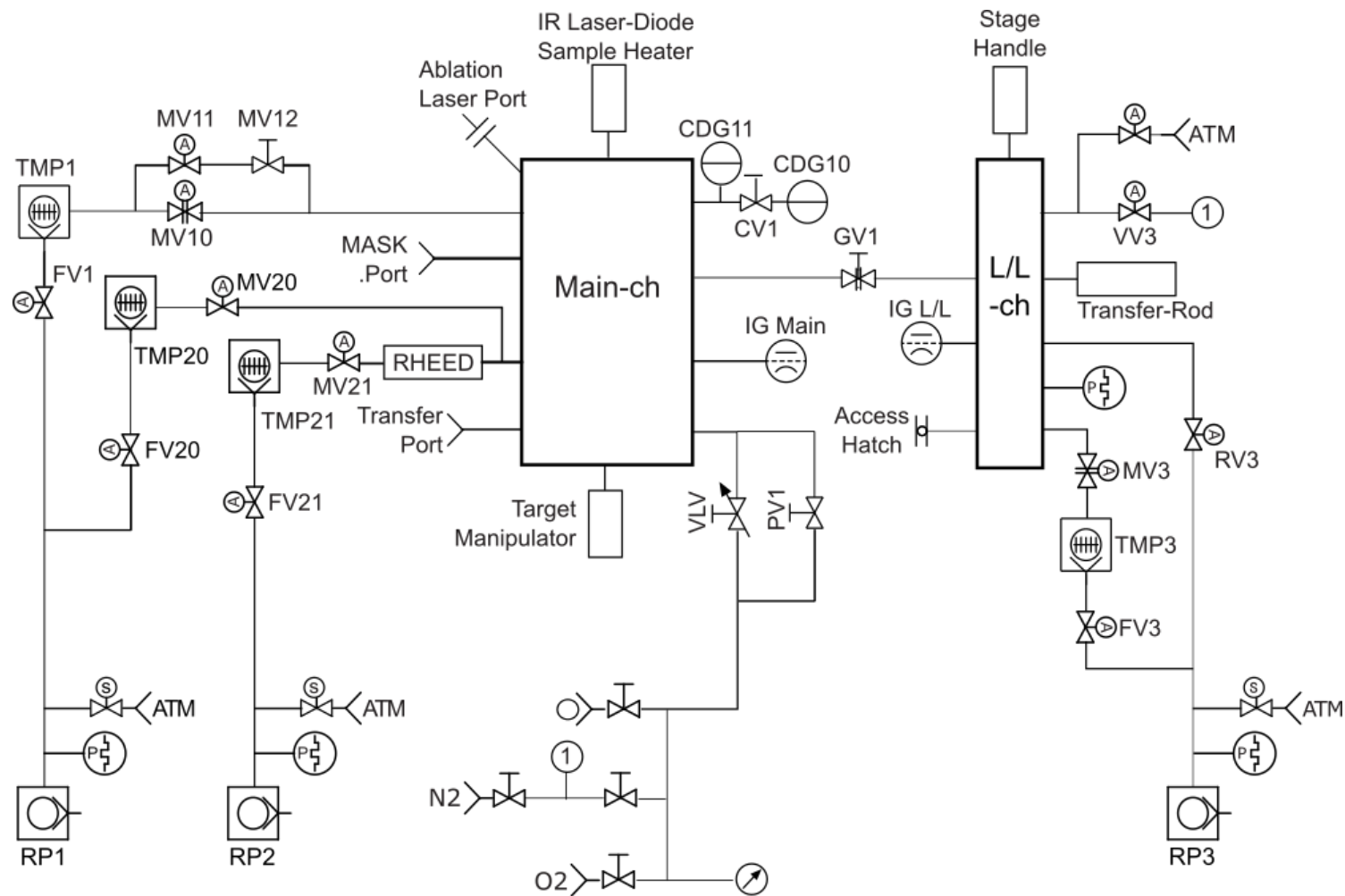


Figure 2: Vacuum diagram of the UVPLD system.

2.1 Main Chamber

The main chamber corresponds to “Main-ch” in Figure 1 and 2. It enables to deposit thin films in wide pressure range by an UHV pumping unit and a process gas inlet. Parts and these functions in the PLD chamber are shown in Figures 3, 4 and Table 1.

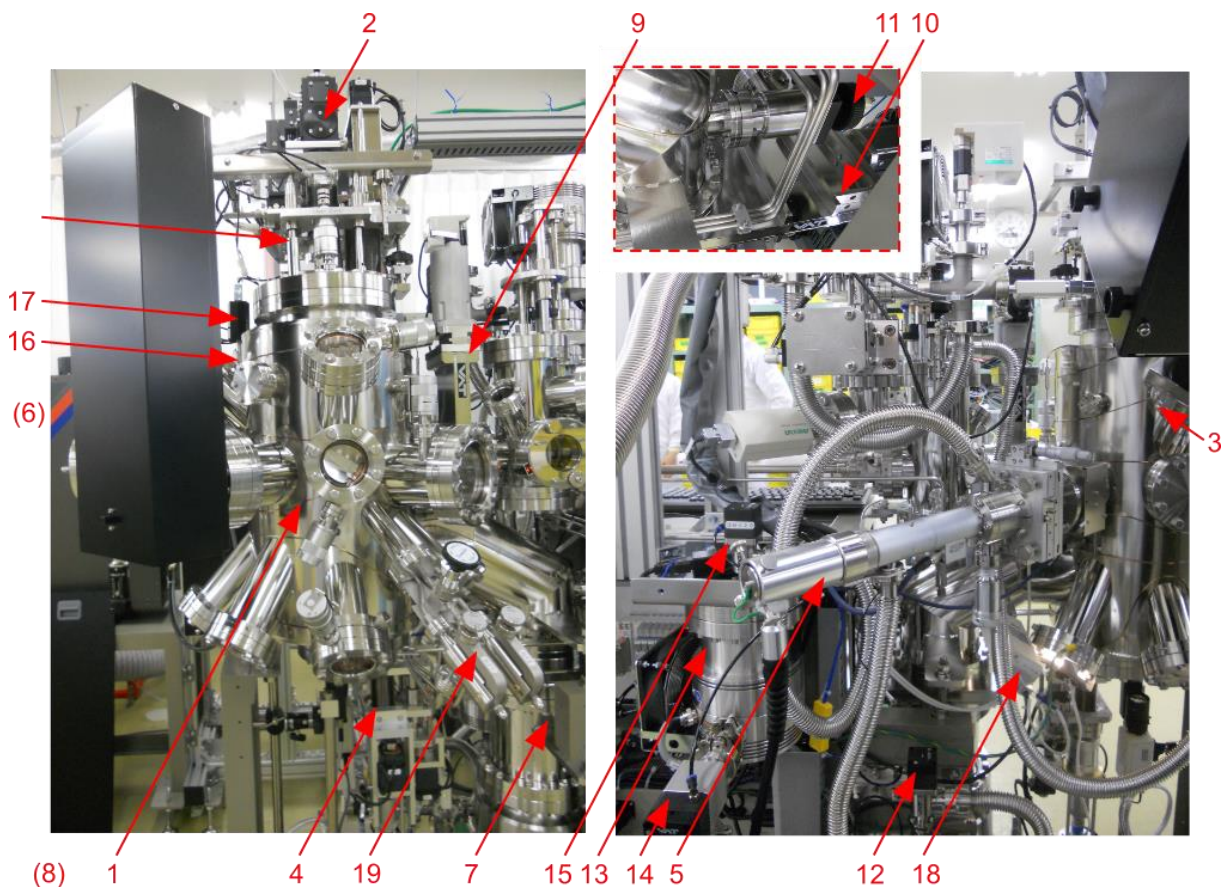


Figure 3: Main chamber (a) front side and (b) rear side.

Table 1: Symbols or names of the parts in the main chamber.

#	Name (see also Figure 3 and 4)	Function
1	Main chamber	
2	Laser-diode substrate-heating unit (see Sec. 2.2)	The unit heats a substrate by irradiation of infrared laser beam to backside of the substrate.
3	Ablation laser port	The port is a window where ablation laser beam goes into the main chamber (vacuum). In the system the ablation laser port has a window rotor to keep clean.
4	Target manipulator (see Sec. 2.3)	The unit selects a source target, adjusts target position, and controls target spinning.

5	RHEED electron-gun (see Sec. 2.5)	The unit is a part of RHEED set to observe surface crystalline structure of deposited film/substrate or to record RHEED oscillation. Double-stage differential pumping is equipped in the PLD system.
6	Combinatorial mask positioning unit (see Sec. 2.4)	The unit makes thin-film pattern or enables dynamic shading during deposition.
7	TMP1: Turbo molecular pump	The pump makes the main chamber UHV.
8	DP1 & 2: Dry pumps	The pump works supplementary for the main chamber pumping line.
9	GV1: Isolation valve (Manual operation)	The valve isolates between the main chamber and the load-lock chamber.
10	MV10: Main valve	The valve connects/cuts main chamber pumping line.
11	MV11: Bypass valve	The valve connects/cuts bypass line.
12	FV1: Fore valve	The valve connects/cuts main chamber pumping line.
13	TMP20, TMP21: Turbo molecular pumps	The pump makes the RHEED e-gun vacuum for high-pressure drive.
14	MV20, MV21: Main valves	The valve connects/cuts RHEED differential pumping line.
15	FV20, FV21: Fore valves	The valve connects/cuts RHEED differential pumping line.
16	CV1: Cut valve	The valve connects/cuts capacitance gauge sensor (CMG: see #17) to the main chamber.
17	CDG10, CDG11: Capacitance Diaphragm gauge	The gauge precisely measures pressure in the main chamber in range of 1.33×10^{-2} Pa to 133 Pa and 13.3 Pa to 133000 Pa under gas-flow condition.
18	IG: Ionization gauge	The gauge measures pressure in the main chamber in UHV range.
19	VLV, PV: Process gas feeder (see Sec. 2.6)	The part introduces process gas like oxygen to the main chamber.
	Viewports	There are some windows with different sizes in the main chamber to see inside during sample/target transfer or deposition.

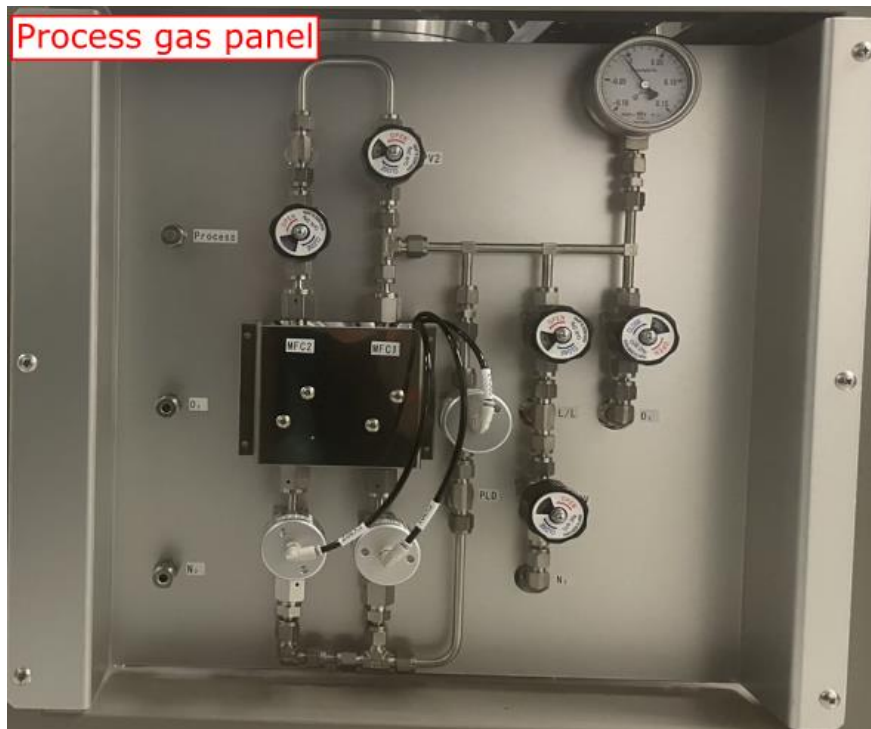


Figure 4: Gass panel.

User can control all pumps and part of valves in pumping line and can monitor vacuum pressure in vacuum controller sub-window of PC software (see item (2) in Sec. 3.3). In pump cases user can start or stop the rotary pumps and the turbo molecular pumps by mouse click in button form. And user can open or close part of the valves. Button operations have order by interlock function.

Caution



(Viewport for observation inside the main chamber)
Bright laser light would scat and come to user's eyes causing injure. Wear a laser-protection goggle when the user wants to observe laser beam path to a source target inside the chamber during deposition.

2.2 Laser-diode substrate heating unit

The substrate heater / manipulator unit is mounted to top side of the PLD chamber. A infrared laser beam path to backside of the substrate is made with the laser optics and motion mechanism inside the chamber. Parts and these functions in the heater unit are shown in Figure 5 and Table 2.

Table 2:Names and functions of the parts in the laser-diode sample heater.

#	Name	Function
1	Laser fiber	The fiber transmits infrared laser light from the laser-diode module.
2	Laser fiber receptacle	The part is the connector of the laser fiber to the heating unit.
3	Cooling fan	The fan prevents overheating of the fiber receptacle and the lens hood.

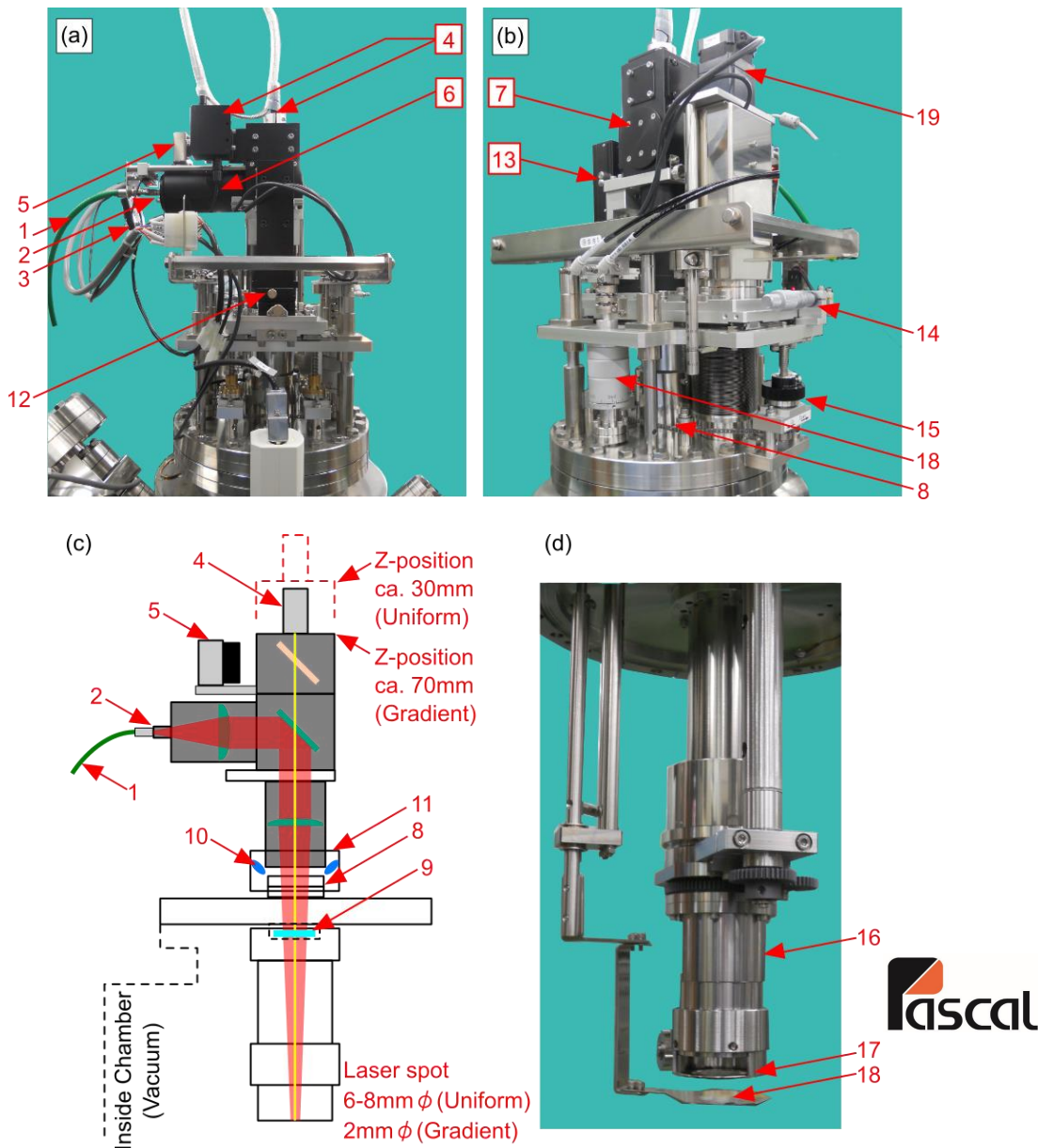


Figure 5:Structure of the laser-diode substrate heating unit.

4	Pyrometer	The sensor measures temperature at backside of a substrate by coaxial view-path with the infrared laser beam path.
5	CCD camera	The camera views backside of the sample or the mask.
6	Lens hood	The part holds two focusing lenses and prevent laser light scattering outside.
7	Mirror box	The box holds split mirror for reflection of laser beam and for transmitting visible light/far IR light for CCD and pyrometer.
8	Laser window (with AR coating)	The window transmit infrared laser light from outside to inside (vacuum) of the main chamber.
9	Glass inner plate	The plate prevents the window contamination by deposition materials.
10	LED light	The light brings clear view for observation by the CCD.
11	Scatter shield	The column covers between the lens hood and the window at top flange.
12	XY axes adjusters for lens hood	The part adjusts the horizontal position of the lens hood.
13	Z axis adjuster for lens hood	The part adjusts the vertical position of the lens hood.
14	XY axes adjusters for substrate stage	The part adjusts the horizontal position of the substrate stage.
15	Z axis adjuster for substrate stage	The part adjusts the vertical position of the substrate stage. It is utilized when substrate holder exchange.
16	Laser shield	The part prevent laser light scattering.
17	Substrate stage	The bottom part of the laser shield. It sets a substrate holder.
18	Substrate shutter	The shutter shade substrate surface.
19	Motor	The motor rotates the substrate stage.

Optical setup with respect to IR laser beam and pyrometer view is also shown in Figure 5(c). User can arrange the laser beam to make appropriate laser spot size at backside of a substrate with 2 plano-convex lenses and changing Z-position of the lens hood.

User can control the laser-diode driving current or substrate temperature on a control panel of PC software (see item 1 in Sec 3-5). And user can control substrate angular position. And user can set endless rotation of the substrate. Both functions are realized PC software (see item 4 in Sec. 3-5).

Caution

	(Laser fiber, Fiber receptacle, Window, Lens hood) Intense infrared laser beam (class 4 and 32/40W) emits from the LD module, goes through the laser fiber, and reaches top side of the main chamber. Take care of damages of related parts by reflecting or scattering laser beam and burning with dust.
	(Fiber receptacle) Leak of laser beam causes overheat near the fiber receptacle or lens hood. Make sure connection of the laser fiber.
	(Motor for sample rotation) Motor with gears or axes partly appear as users easily touch. Do not make users bit by the parts.

3	Rotary motion feedthrough for target spin	The parts transfer rotary motion of target spinning into inside (vacuum) of the main chamber.
4	Servo motor for target spin	The motor works for target spin.
5	Z-axis adjuster	The part adjusts vertical position of the target stage. It is utilized when target setting or detaching.
6	Stepping motor for Z-axis <i>(option, not included)</i>	The motor works for Z-axis adjust.
7	Shield	The part prevent that ablated material from a target contaminates other target material during deposition.
8	Target stage	The part sets target holders by attaching cone-shaped convex bottoms to cone-shaped concave tops of target axes.
9	Viewport	The windows enables to see front-side of the sample.

Ablation laser beam goes to irradiate target surface in blue line in Figure 6. And user can measure laser power at opposite-side window when the user set “Clear” position of the target stage. The same position “Clear” allows to view sample surface from the bottom-side window like the green dotted line in Figure 6. User can control all target manipulation of choosing target materials or setting operation mode of target twist and spin on above panel of PC software (see item 3 in Sec. 3.5).

Caution

	(Target manipulator in maintenance) The unit is heavy over than 20kgf. In maintenance mount /dismount of the unit should be done by plural persons with helping each other.
	(Rotary motion feedthroughs, Motors, axis adjusters) Rotary/linear motions with gears or axes partly appear as users easily touch. Do not make users bit by the parts.

2.4 Combinatorial mask positioning unit

There is a mask-positioning units in the main chamber as shown in Figure 7. It is the “long mask unit” with about 160 mm stroke and trapezoidal hole-patterns those can make composition-spread thin-film pattern like triangular phase diagram. The mask positioning units can be manually controlled by handy console and precisely controlled by computer software (see item 5 in Sec 3.5).

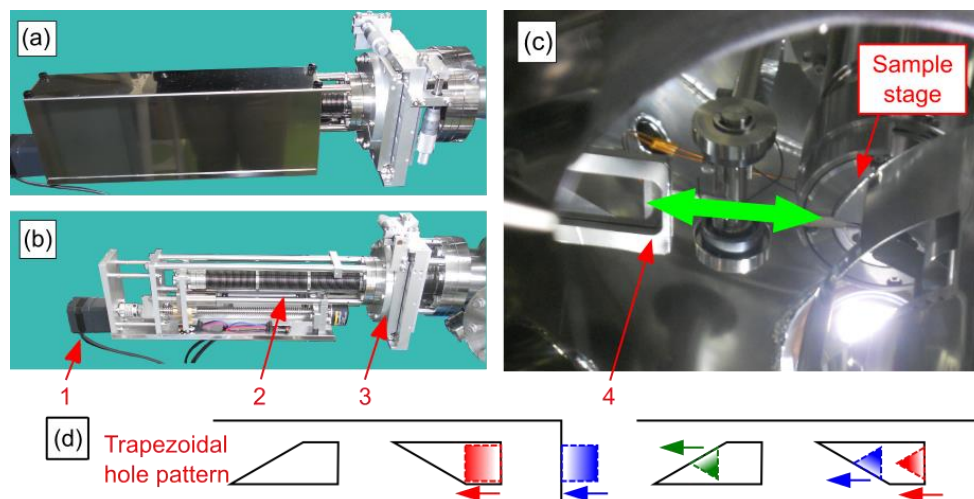


Figure 7:Combinatorial mask positioning unit.

Table 4:Names and functions of the parts in the combinatorial mask positioning unit.

#	Name	Function
1	Stepping motor	The motor is utilized for long-mask positioning with sensor (limit switch and rotary encoder).
2	Linear motion feedthrough	The part supports linear motion of the mask in vacuum. Its stroke is about 160mm.
3	XY-axis adjuster	The part adjusts horizontal/vertical positions of the mask.
4	Mask plate	The mask plate moves beneath the sample and make thin-film pattern. Trapezoidal mask pattern is provided at present.

Caution



(Linear/rotary motion feedthroughs, Motors, axis adjusters)
Linear/rotary motions with gears or axes partly appear as users easily touch. Do not make users bit by the parts.

2.5 Reflection high-energy electron diffraction (RHEED) unit

The reflection high-energy electron diffraction (RHEED) unit is a analytic tool to observe surface crystalline structure or a real-time monitoring tool during layer-by-layer crystalline film growth. In the PLD system the RHEED electron gun has a option: double-stage differential pumping unit for high-pressure RHEED pattern observation in addition to the close-up fluorescent screen. Outline of the RHEED observation set-up is shown in Figure 8 and Table 5.

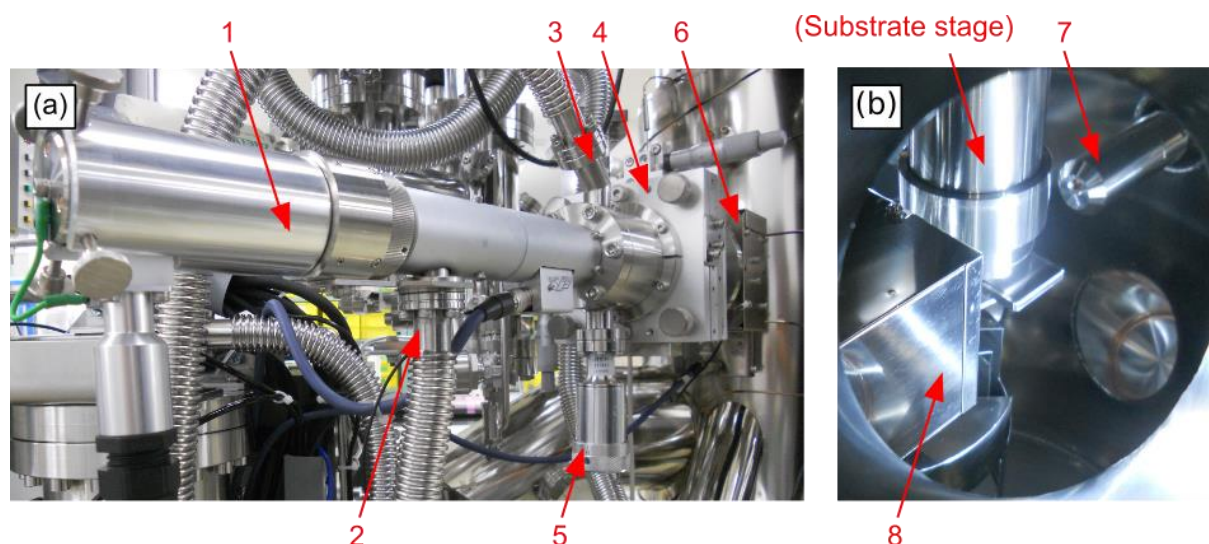


Figure 8:RHEED electron gun with double-stage differential pumping line and scanning coil.

Table 5:Names and functions of the RHEED electron gun.

#	Name	Function
1	RHEED electron gun	<i>Detail description about operation and maintenance are shown in attached documents.</i>
2	First differential pumping port	Vacuum pumping through the port is for preventing filament burn-out.
3	Second differential pumping port	Vacuum pumping through the port is for high-pressure RHEED electron-gun operation in addition to the above pumping port.
4	XY/Tilt electron-gun stage	The stage adjusts the electron-gun mounting position and following incident electron beam path without deflection.
5	Emission aperture exchange knob	The part exchanges the hole, which is at the end of the electron-beam guide pipe, between a large diameter and a small.
6	Scanning coil	The secondary deflection coil makes psudo in-plane parallel e-beam bundle towards substrate surface.
7	Electron -beam guide pipe	The part guides the electron beam close to the sample, and prevents permeation of the process

		gases into the electron gun.
8	Close-up fluorescent screen and shield	The screen shows diffraction pattern effectively with avoiding electron-beam scattering by a large amount of process gasses.



Figure 9:RHEED fluorescent screen and camera hood.

Caution

	(RHEED operation environment) X-ray is generated when an electron beam is irradiated to a substrate material with RHEED electron gun. Keep away from viewing ports which have no protection.
	(HV inside RHEED electron gun) High voltage up to 30kV is applied at apart of RHEED electron gun. Do not touch the electric conduction parts.

2.6 Process gas panel/process gas inlet

The process gas panel is a subpart of the PLD chamber (see Figure 10). It can supply oxygen, argon, and nitrogen gases to the main chamber and the load-lock chamber after connection of primary side. All gases can be introduced to the PLD chamber with flow control through variable-leak valve (VLV) or process valve (PV).

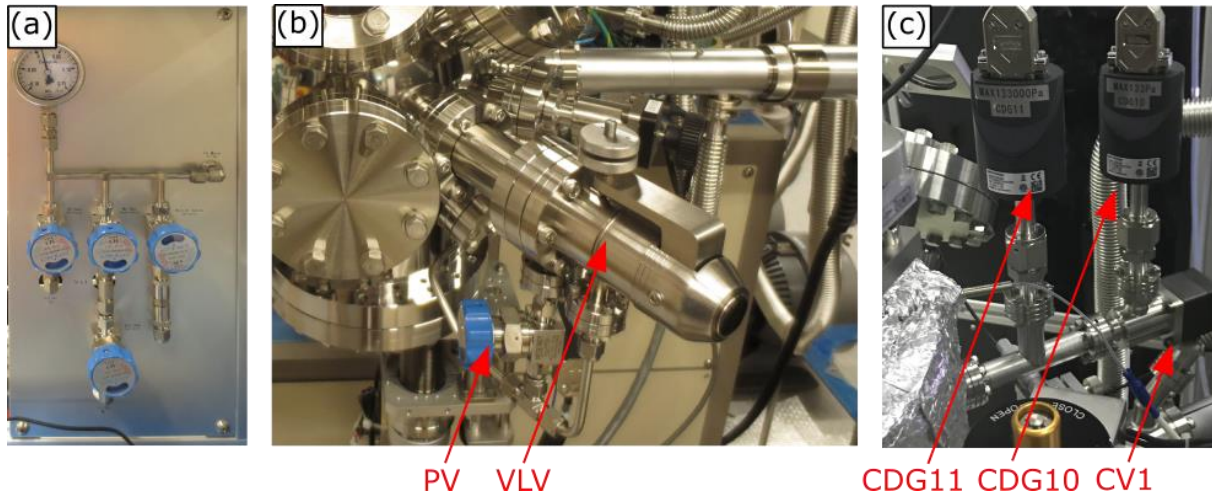


Figure 10: Process gas feeder (a) gas panel, (b) inlet valves and (c) capacitance gauge.

2.7 Load-lock chamber

Parts and these functions in the load-lock chamber are shown in Figure 11 and Table 6.

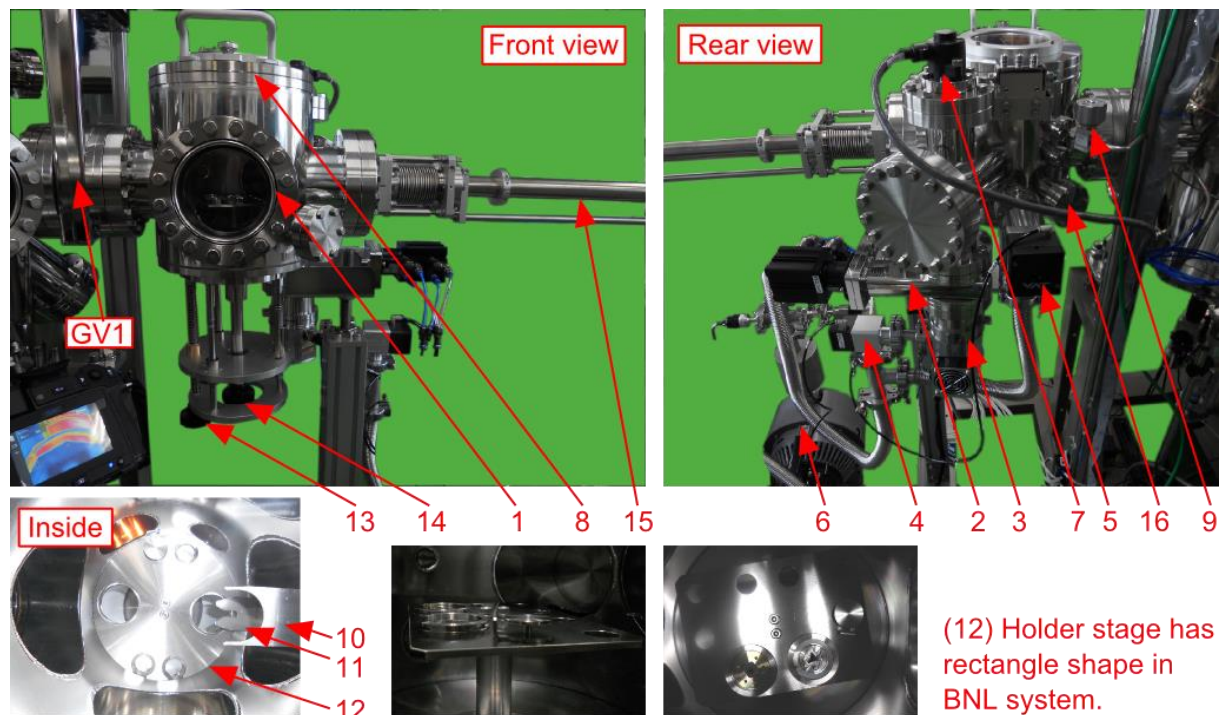


Figure 11:Parts and in load-lock chamber.

Table 6:Names and functions of the parts in the load-lock chamber.

#	Name	Function
1	Load lock (L/L) chamber	The chamber make vacuum in range of HV to ATM for sample/target exchange.
2	MV3: Main valve	The valve connects/cuts L/L chamber pumping line.
3	TMP3: Turbo molecular pump	The pump makes HV for sample/target exchange between the main chamber and L/L chamber.
4	FV3: Fore valve	The valve connects/cuts L/L chamber pumping line.
5	RV3: Rough valve	The valve connects/cuts L/L chamber bypass pumping line.
6	DP3: Dry pump	The pump works in main pumping line.
7	IG3: Ionization gauge	The gauge measures pressure in the main chamber in UHV region.
8	Access hatch:	The door opens L/L chamber to exchange samples/targets.
9	VV1	The valve feeds vent N2 gas into L/L chamber.
10	Sample transfer fork	Upper large fork catches a sample holder.
11	Target transfer fork	Lower small fork catches a target holder.

12	Holder stage	The stage sets two sample holders and two target holders.
13 14	Manipulation handle of holder stage	Two handles below the L/L chamber adjust the holder stage position with up/down motion and rotation to catch/release these holders.
15	Transfer rod	Knob of the transfer-rod pushes in/pulls out the forks to the main chamber.
16	Reserved port	The port is used for roughing at the ablation laser port with the valve MV5.

User can control all pumps and part of valves in pumping line and can monitor vacuum pressure in vacuum controller window of PC software (see item 2 in Section 3.3). In pump case user can start or stop the rotary pumps and the turbo molecular pumps by mouse click in button form. And user can open or close part of the valves. Button operations have order by interlock function. In the PLD system case automatic vent and pumping function with one-click operation are equipped.

2.8 Control Panel / PC software

The control panel near vacuum chamber provides fundamental operation of the system. Parts and these functions are shown in Fig. 12 and Table 7.

Table 7: Names and functions of the target manipulator.

#	Name	Function
1	Computer	The computer includes IR laser deposition control software.
2	Handy console	The controller enables to move motorized motion like sample rotation, target manipulation, mask positioning manually.
3	Pre-annealing heater controller	The controller sets pre-annealing temperature.
4	Pre-annealing heater timer	The timer sets pre-annealing period.
5	Baking heater controller	The controller sets baking temperature.
6	Speed controller	The controller sets spinning speed of source target.
7	Emergency stop button (EMO)	User can shutdown the whole system immediately in emergency cases.
8	RHEED scanning coil driver	The driver unit controls second deflection coil current for parallel scanning of electron beam (see optional manual).
9	RHEED HV power supply	The unit controls RHEED electron beam emission (see optional manual).
10	LD controller/power supply	The unit contains a laser-diode module and its power supply for sample heating.
11	Vacuum gauge controller	The controller shows vacuum pressure values and sets measurement parameter.
12	Pre-annealing heater power supply	The unit controls pre-annealing heater power.
13	TMP controller (TMP1)	The controller drives TMP for the main chamber.
14	TMP controller (TMP20)	The controller drives TMP for differential pumping line of the RHEED electron gun.
15	TMP controller (TMP21) <i>(option, not included)</i>	The controller drives TMP for differential pumping line of the RHEED electron gun.
16	TMP controller (TMP3)	The controller drives TMP for the L/L chamber.
17	Main switch	
17	Main switch	

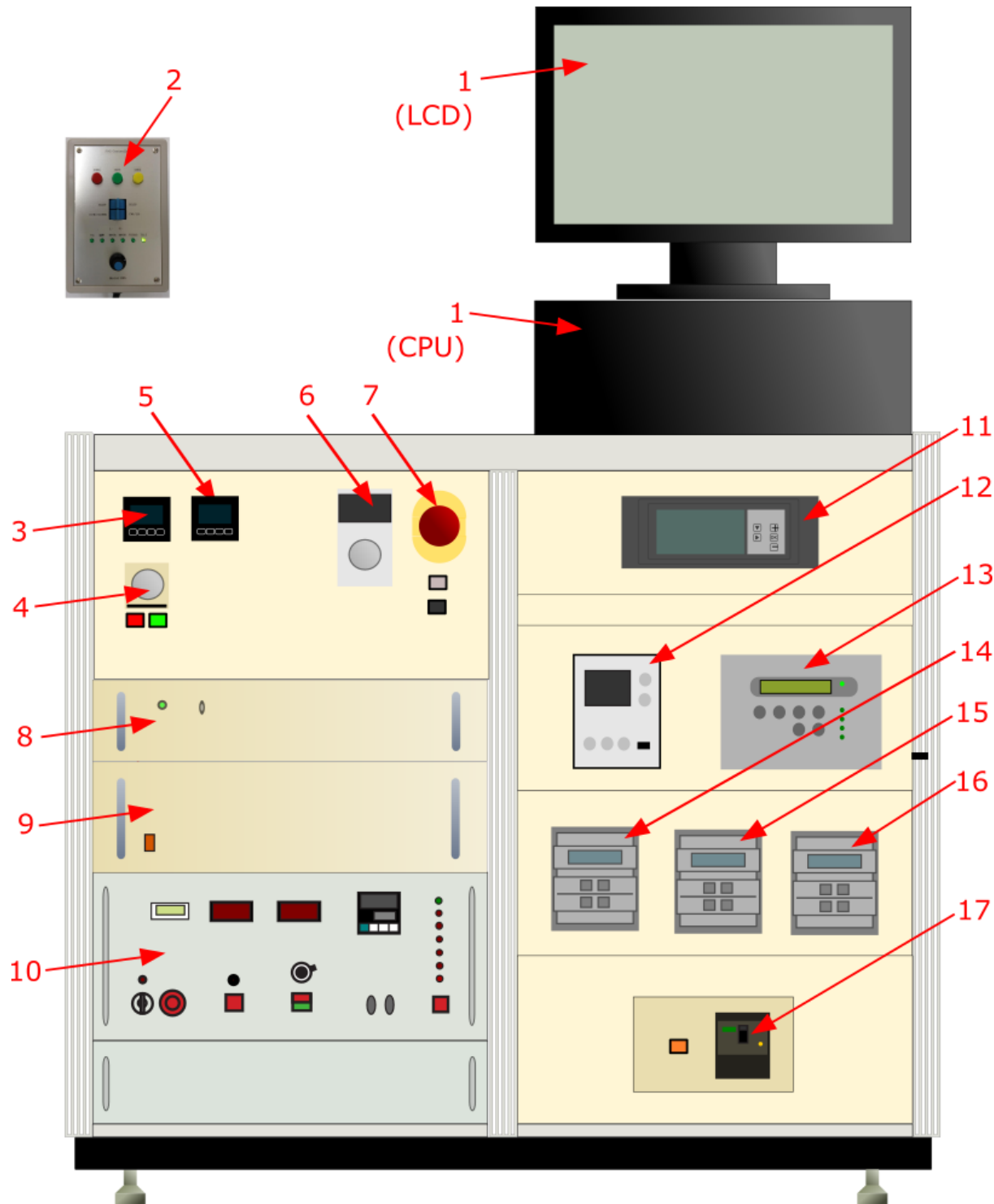


Figure 12:Control panel of the PLD system.

2.9 Ablation laser and optical box

General optics setup of the UV ablation laser for the PLD system is shown in Figure 13. See also manual for the ablation laser model COMPex.

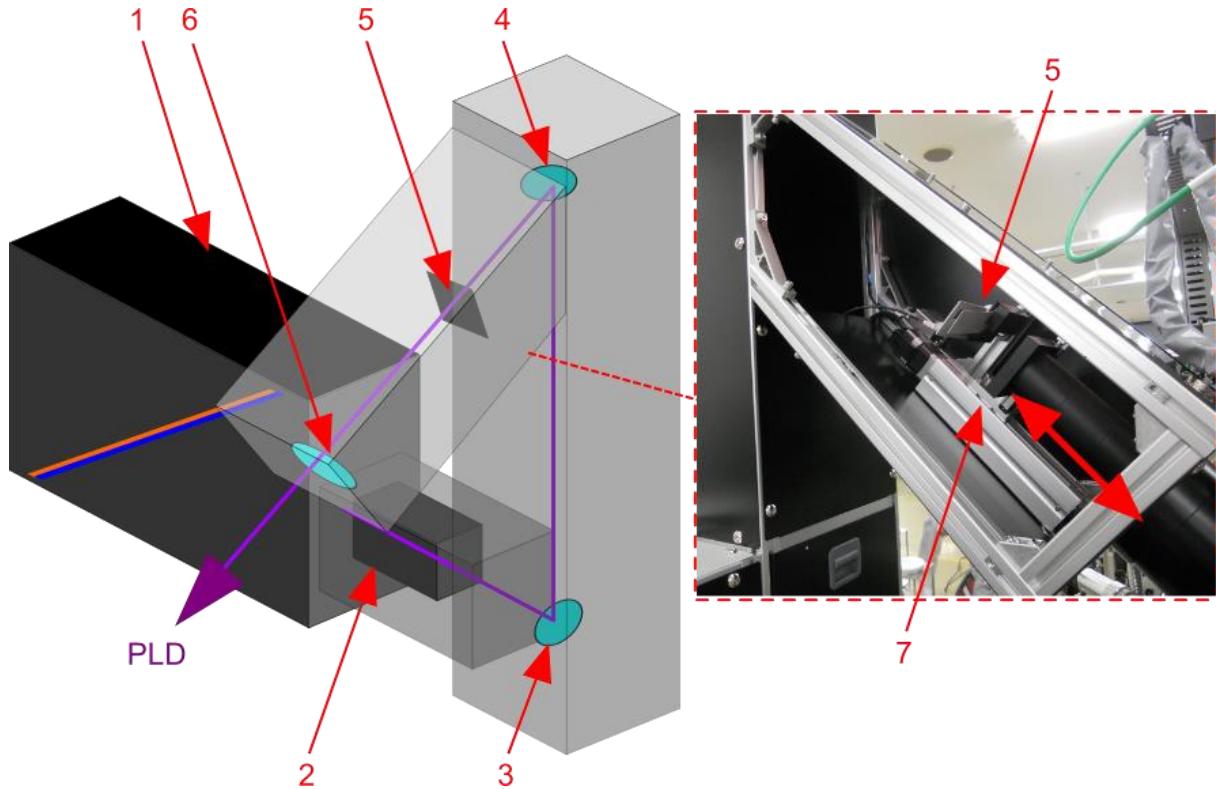


Figure 13: Ablation laser and optical box in the UVPLD system.

Table 8: Names and functions of the ablation laser and the optical box.

#	Name	Function
1	UV ablation laser	COMPex is set for the PLD system.
2	Attenuator <i>(option, not included)</i>	The optics varies UV ablation laser intensity.
3	Reflection mirror #1	The mirror has reflection angle of 45 degree for KrF excimer laser light ($\lambda=248$ nm).
4	Reflection mirror #2	The mirror has reflection angle of 22.5 degree for KrF excimer laser light ($\lambda=248$ nm).
5	Gate shutter	
6	Focusing lens	The lens has focal length of 500 mm as nominal.
7	Lens stage	The stage varies focal length towards the target surface

The ablation laser port has internal quartz plate which can be rotated and be exchanged to prevent unintentional coating of deposition material during long-time use. Structure of the ablation laser port is

shown in Figure 14.

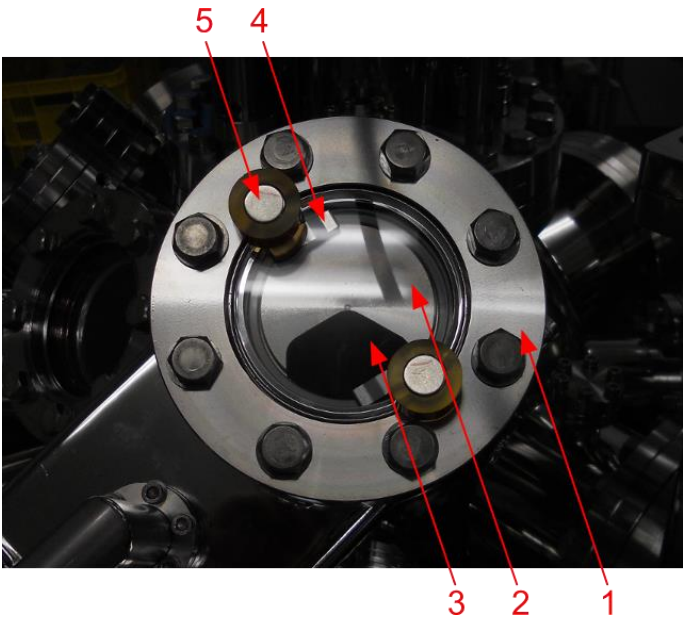


Figure 14:Structure of the ablation laser port.

Table 9:Names and functions of the ablation laser port.

#	Name	Function
1	Quartz viewport	Flange size: ICF114 (CF4.5").
2	Quartz inner plate	The plate can be rotated to prevent contamination.
3	Laser beam window	The window opens partly to use the inner quartz plate (#2) for long time.
4	Inner magnet	
5	Outer magnet	

3 Software guide of the PLD systemf

3.1 Outline of the control software

This software provides easy remote operations of vacuum pumping, source target manipulation, ablation laser oscillation, mask positioning, and sample rotation for PLD/laser MBE (LMBE) film fabrication process. The software unifies above remote operation factors in one main application window for systematic control. And some process parameters can be seen in real-time monitor panel.

This software has two remote control modes. One is a manual operation mode in the software. In manual mode all PLD process parameters can be changed by adjusting/inputting operation in the main window. The other is a “script” operation mode. The script operation mode is automatic control of deposition sequence to realize combinatorial thin-film sample fabrication. A script file is a text base document written with easy-understanding keywords of “commands” or “construction sentences” like repetition control. And the script file can be edited by script editing/executing sub-window or other editor software. So the script operation mode can accomplish complex PLD/LMBE procedure like multilayered structure, tiling systematic synthesized pixel samples on a substrate, composition-spread on a substrate, or others. It is the powerful tool for sample fabrication and high-speed screening of process condition.

All windows/screens of the software are constructed with GUI (Graphical User Interface) by Lab View. Most of the deposition parameters and deposition process can be changed/controlled by mouse operation.

3.2 Caution

This software realizes systematic control, real-time monitoring, and automatic control of deposition sequence. For above purpose the software always works tightly in data communication to PLD/LMBE process units and data processing though the software is an application-software in Microsoft Windows operation system (OS) with PC and extension boards. So PASCAL would like users not to do following tasks and operations.

- (1) Installation of other application-software in the PC. In particular installation and execution of other software which wastes much Windows OS resources.
- (2) Mounting other extension boards into the PC.
- (3) Modifying PC hardware construction out of maker/factory (PASCAL) design.
- (4) Changing files' (folders') name or files' (folders') location in the PC out of factory (PASCAL) design.



3.3 Startup windows/Screen layout

There are two startup windows when user run the PLD controller software. Screen layout of each window is shown as follows.

(1) Main window

The PLD controller software shows following application-window (see Figure 15). Functions on buttons or control panels are described in Table 10.

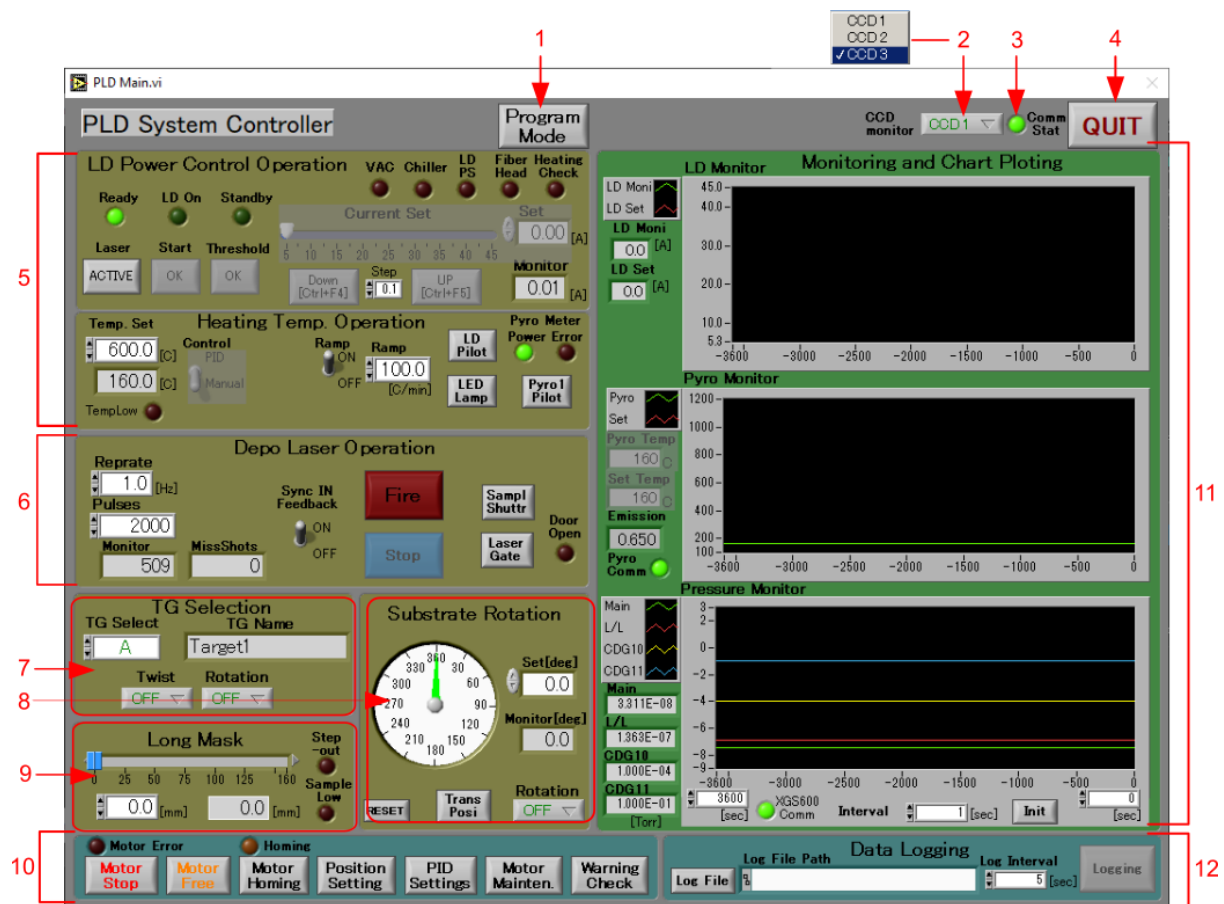


Figure 15: Main window of the PLD control software.

Table 10:Panel names and functions of the main window.

#	Name	Functions / Corresponding local operations
1	Script sub-window button	Script sub-window appears when click. <i>(No local operation.)</i>
2	CCD selector button	Camera image changes among bottom-mount CCD, side-mount CCD, or RHEED camera when choosing items of the button. <i>(No local operation.)</i>
3	Comm Status indicator	This indicator lights when signal communication is confirmed between PC and PLC. <i>(No local operation.)</i>
4	Quit button	Quit the software when click. <i>(No local operation.)</i>
5	Substrate-heating control panel	This part controls substrate temperature in laser-diode driving current mode or PID feedback-control mode with pyrometer. <i>(Local operation in control panel of laser diode power supply.)</i>
6	Ablation-laser oscillation control panel	This part controls ablation-laser oscillation. <i>(Local operation in ablation-laser.)</i>
7	Source target manipulation panel	This part manipulates source target. <i>(Local operation with handy consol.)</i>
8	Substrate rotation control panel	This part controls substrate rotation. <i>(Local operation with handy consol.)</i>
9	Mask positioning control panel	This part controls position of combinatorial mask. <i>(Local operation with handy consol.)</i>
10	Supplemental operation panel	This part does extra motorized operations like emergency stop or homing. And the part sets some parameters for above functions. <i>(Local operation with handy consol for motorized operations.)</i>
11	Monitor panel	This part shows status of the PLD system. <i>(No local operation.)</i>
12	Logging panel	This part records status of the PLD system to a PC file. <i>(No local operation.)</i>

(2) Vacuum control window

Vacuum controller window shows the simple vacuum diagram with control buttons in accordance to installed hardware including pumps, valves, gauges and others. Each part works as follows.

Table 11:Panel names and functions of the vacuum control window.

#	Name	Functions / Corresponding local operations
1	Vacuum pumping operation part	User can turn on/off pumps and open/close (pneumatic) valves by mouse-click on the panel. And the part shows vacuum pressure. <i>(No local operation.)</i>
2	Chamber baking controller	The part control chamber baking heater with turn-off time. <i>(No local operation.)</i>
3	Alarm indicator	The part lights/sounds an alarm and informs troubles of the system in message box. <i>(No local operation.)</i>

(3)

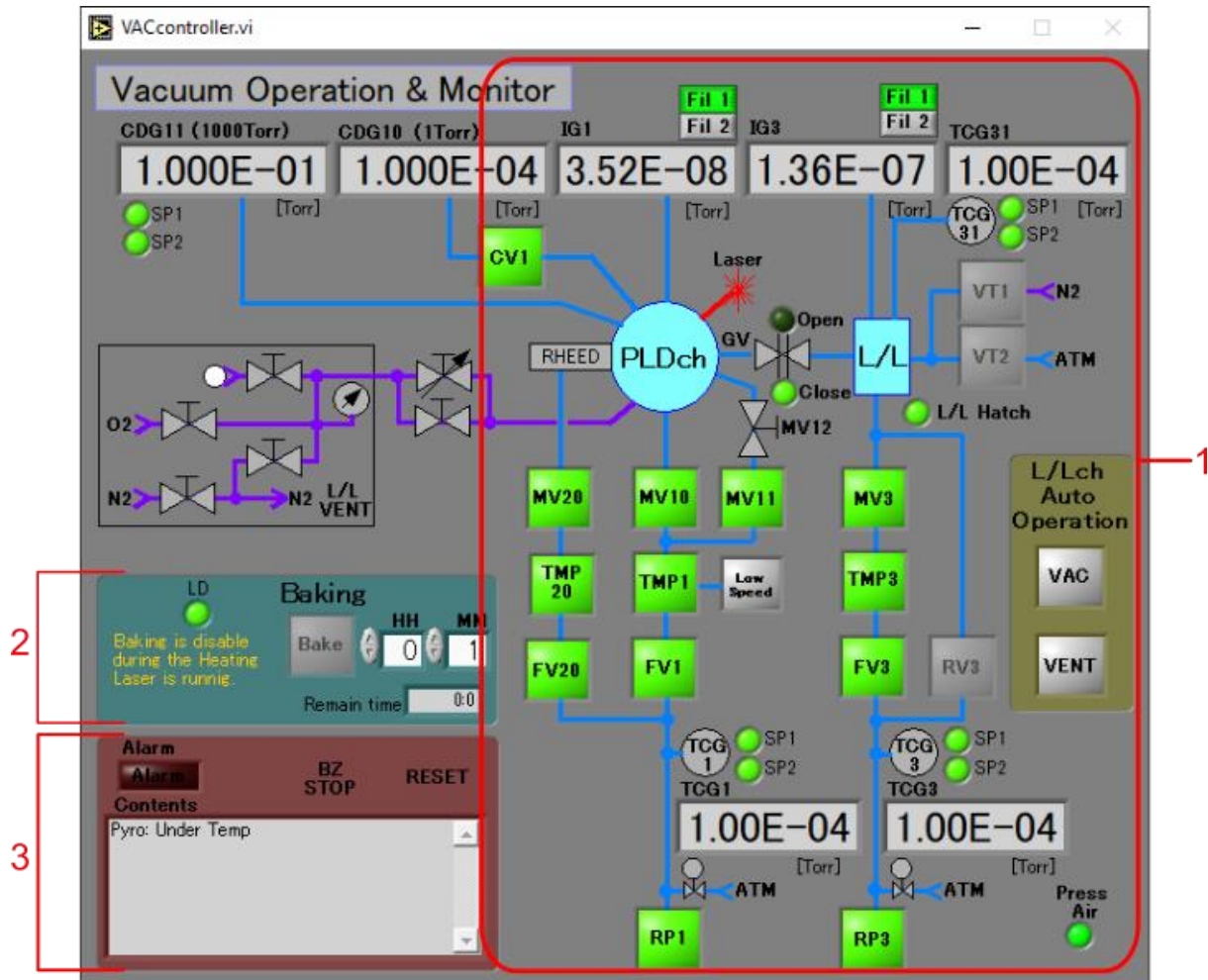


Figure 16: Vacuum controller and monitor window.

3.4 Sub-windows

Sub-windows appear by button operation on main window. Each function is shown as follows.

(1) Supplemental function window

A sub-window, which shows and adjusts supplemental function parameters, appears by button operation on the control panel (see the item in Figure 1-10). Detail explanation is written in chapter 4.

(2) Script edit/run sub-window

A sub-window appears by button operation on main window.

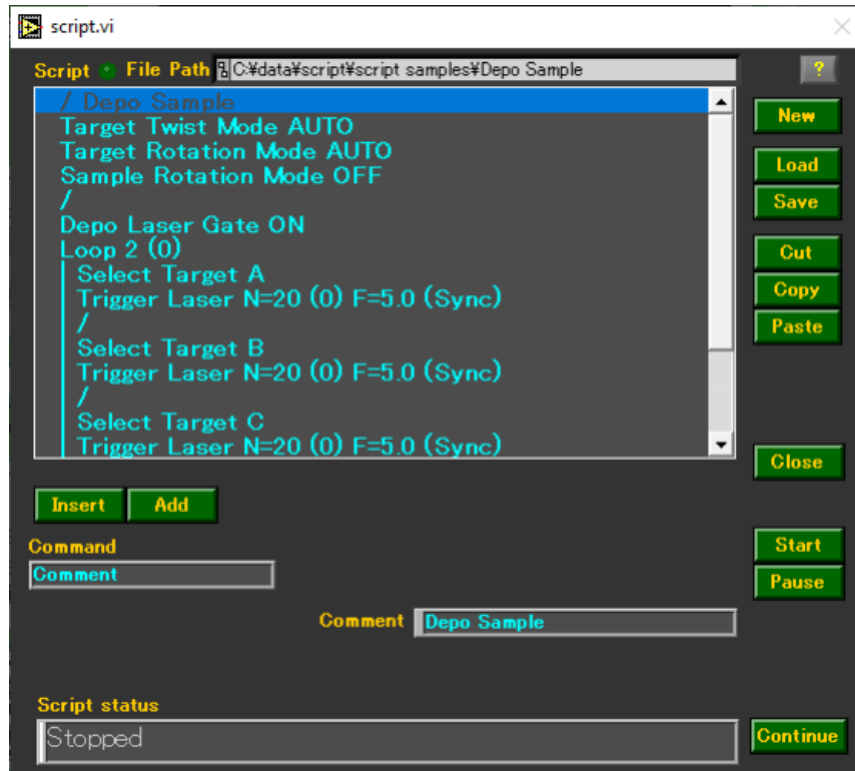


Figure 17:Script editing/executing sub-window.

Fundamental window set (screen-set), two startup windows and sub-windows, provides most of PLD (laser MBE) process in two control modes.

Manual (remote) control of each control factor for PLD (laser MBE) process

All parameters for PLD process are set at control panels in the main window. And operations are carried out in some buttons. Detail functions and operations are shown in chapter 3.

Automatic control of PLD (laser MBE) process by script

Script, which is a programmable sequence for automatic control of PLD process, is edited or operated in the script-sub-window. Script controls the process with preset parameters in the main window or parameters set in the script file. Operation of the script sub-window is shown in chapter 5 in operation edition. And script command reference is also described in chapter 6 of the operation edition.

3.5 Control/monitor panel in main window

(1) Substrate-heating by infrared laser light with PID temperature control

This part in the main window controls substrate temperature by operating laser-diode driving-current. Upper side of the part has switches to turn laser-diode and pilot laser on/off or driving-current adjusters corresponding to the control panel in laser-diode power supply.

Lower side of the part has substrate temperature controller in PID mode when pyrometer works with data communication. Names and functions of the part is as follows.

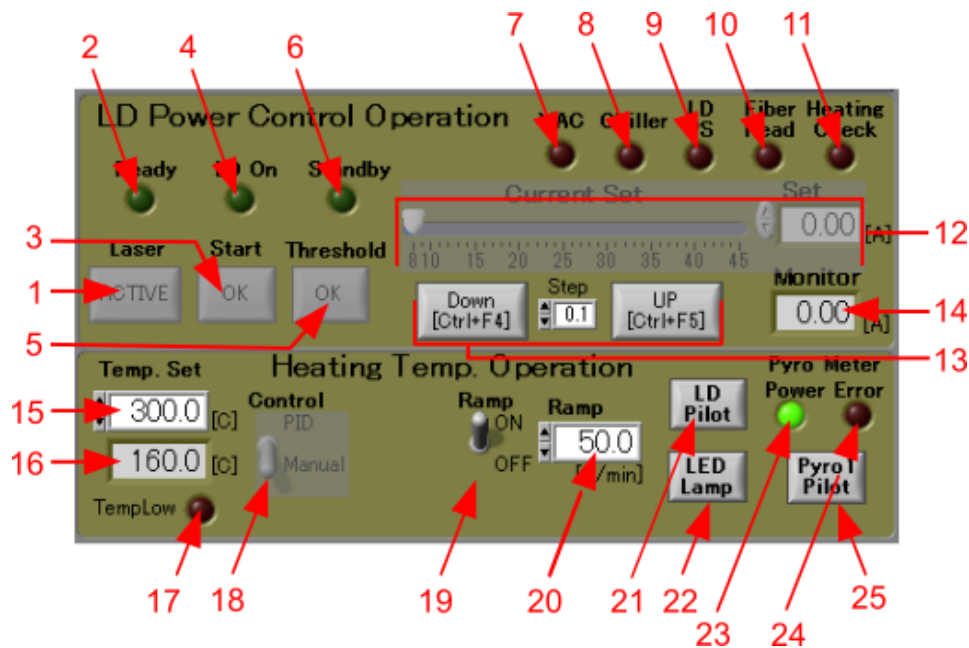


Figure 18:Laser-diode substrate heating control panel.

Table 12:Functions in laser-diode substrate-heating control panel.

#	Names / Functions	Corresponding local operations
1	Laser switch: this switch permits or forbids operating switches or adjusters in the part.	No local operation.
2	Ready indicator: it lights in green when “Active” at laser switch (#1) is selected.	No local indicator.
3	Start switch: this switch turns the laser-diode module on/off.	Local operation with LD button on front panel of laser-diode power supply.
4	LD on indicator: it lights in green when “ON” at start switch (#3) is selected.	Local indicator is on front panel of laser-diode power supply.
5	Threshold switch: this switch warms the laser-diode module up at threshold current for oscillation.	No local operation.

6	Stand-by indicator: it lights in green when warm-up process (#5) is successfully done.	No local indicator.
7	VAC alarm: the notice of red light means illegal vacuum condition during sample heating.	No local indicator.
8	Chiller alarm: the notice of red light means troubles in chiller.	No local indicator.
9	Power supply alarm: the notice of red light means troubles in laser-diode power supply.	Local indicator is on front panel of laser-diode power supply.
10	Fiber head alarm: the notice of red light means troubles in laser-fiber like disconnection.	No local indicator.
11	Temp check: the notice of red light means late/no response from pyrometer due to large thermal inertia at beginning of substrate heating.	No local indicator.
12	LD driving current adjuster: the slider varies the LD driving current. And the right box sets the LD driving current by numerical input.	Local operation with the adjuster knob on front panel of laser-diode power supply.
13	LD driving current adjuster: the UP/DOWN buttons increase or decrease the LD driving current step by step. The box at middle is step value setter.	Local operation with the adjuster knob on front panel of laser-diode power supply.
14	LD current monitor: the box shows present value of LD driving current.	Local indicator is on front panel of laser-diode power supply.
15	Substrate temperature setting box: the box sets the sample temperature and shows it.	No local operation.
16	Substrate temperature monitor: the monitor shows the current sample temperature.	No local operation.
17	Temp low alarm indicator: it lights in red when temperature is too low to control.	No local operation.
18	Temperature control mode selector: this switch sets temperature control mode, PID or manual.	No local operation.
19	Ramp rate set switch: this switch set the ramp rate or not. It is valid in PID mode.	No local operation.
20	Ramp rate setting box: the box sets the ramp rate and shows it. The box is valid in PID mode.	No local operation.
21	LD Pilot lamp switch: this switch turns pilot lamp on/off.	Local operation with Pilot button on front panel of laser-diode power supply.
22	LED lamp switch: this switch turns LED lamp to view the substrate backside clearly.	No local operation.
23	Pyrometer power on indicator: the indicator lights when pyrometer is ready to operate.	No local operation.
24	Pyro Error alarm indicator: the indicator lights when pyrometer cannot measure sample temperature.	No local operation.
25	Pilot lamp switch (pyrometer #1): this switch turns pilot lamp on/off.	No local operation.

(2) Ablation laser oscillation

This part in the main window controls ablation laser oscillation. The figure and the explanation below is assumed to be an excimer laser as an example.

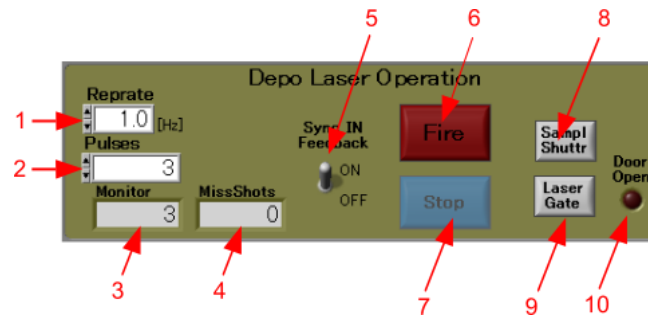


Figure 19:Ablation laser oscillation control panel.

Table 13:Functions in ablation laser oscillation control panel.

#	Names / Functions	Corresponding local operations
1	Repetition setting box: the box sets the repetition rate of the laser pulse oscillation and shows it.	Local operation with console of the ablation laser.
2	Pulses setting box: the box sets the shot number of the ablation laser pulse and shows it.	Local operation with console of the ablation laser.
3	Pulse counter: the box shows the present shot number of the ablation laser pulse.	Local operation with console of the ablation laser.
4	Miss-shots counter: the box shows the miss-shot number to the trigger signal.	
5	Sync switch: the switch sets synchronized controls to ablation laser oscillation.	No local operation.
6	Fire button: the ablation laser oscillation begins when pushing the button.	Local operation with console of the ablation laser.
7	Stop button: the ablation laser oscillation ceases when pushing the button.	Local operation with console of the ablation laser.
8	Sample shutter switch: the switch controls opening/closing the sample (substrate) shutter.	No local indicator.
9	Laser gate shutter switch: the switch controls opening/closing the laser gate.	No local indicator.
10	Door open indicator: the inter-lock alarm means unexpected opening of the optical box.	No local indicator.

(3) Source target manipulation

This part in the main window controls source target manipulation.

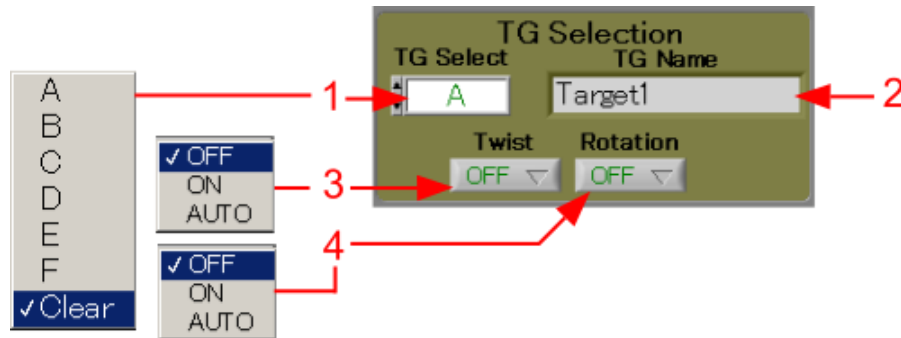


Figure 20:Source target manipulation panel.

Table 14:Functions in source target manipulation panel.

#	Names / Functions	Corresponding local operations
1	Target selector: the selector by operating UP/DOWN cursors or a pull-down list sets a target by indexes A to F and Clear.	Local operation with handy console.
2	Target name display box: the box shows the target name corresponding to the indexes.	No local operation.
3	Target twist mode selector: the pull-down list sets the target twist mode, OFF, ON, or AUTO. In AUTO mode target twist works synchronously with ablation laser oscillation.	No local operation.
4	Target spin mode selector: the pull-down list sets the target spin mode, OFF, ON, or AUTO. In AUTO mode target spin works synchronously with ablation laser oscillation.	No local operation.
5	Target height-position adjuster: the slider sets the height position of the source target.	Local operation with handy console.
6	Target height-position setting box: the box sets the height position of the source target by numerical input.	Local operation with handy console.
7	Target height-position monitor: the monitor shows current height-position of the source target.	No indicator in local operation part.

(4) Substrate rotation

This part in the main window controls angle position of substrate.

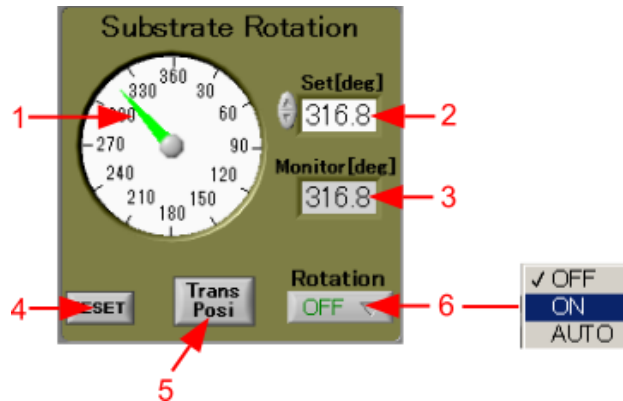


Figure 21:Substrate rotation control panel.

Table 15:Functions in substrate rotation control panel.

#	Names / Functions	Corresponding local operations
1	Angle adjuster: the circle gauge adjuster sets the angle position of the substrate manually.	Local operation with handy console.
2	Angle setting box: the box sets the angle position of the substrate by numerical input.	Local operation with handy console.
3	Angle monitor: the box shows current angle position of the substrate.	No indicator in local operation part.
4	Position reset button: the button set the present angle position to zero.	No local operation.
5	Transfer position button: substrate stage rotates to proper angle position for transfer between main chamber and load-lock chamber.	No local operation.
6	Substrate rotation selector: the pull-down list sets endless substrate rotation mode. There are OFF, ON, or AUTO. In AUTO mode substrate rotates synchronously with ablation laser oscillation.	No local operation.

(5) Mask positioning

This part in the main window controls position of combinatorial masks.

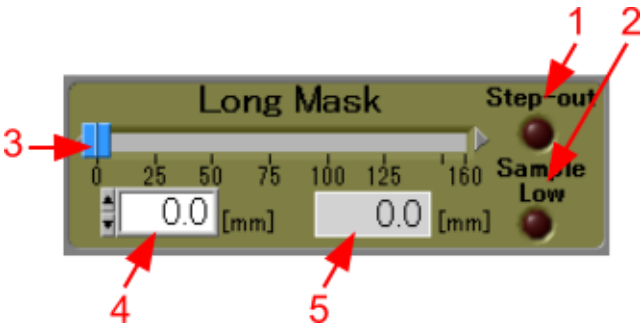


Figure 22:Combinatorial mask positioning panel.

Table 16:Functions in combinatorial mask positioning panel.

#	Names / Functions	Corresponding local operations
1	Step out indicator: Red lamp mans out of control in motorized motion.	No indicator in local operation part.
2	Sample low indicator: Red lamp notices that the sample position is too low in danger of mask collision.	No indicator in local operation part.
3	Long mask adjuster: the slider adjuster sets the position of the long mask manually.	Local operation with handy console.
4	Long mask setting box: the box sets the position of the long mask by numerical input.	Local operation with handy console.
5	Long mask monitor: the monitor shows current position of the long mask.	No indicator in local operation part.

(6) Status monitor and logging

The right part in the main window has chart monitor of laser-diode driving-current, substrate temperature, and chamber pressure.

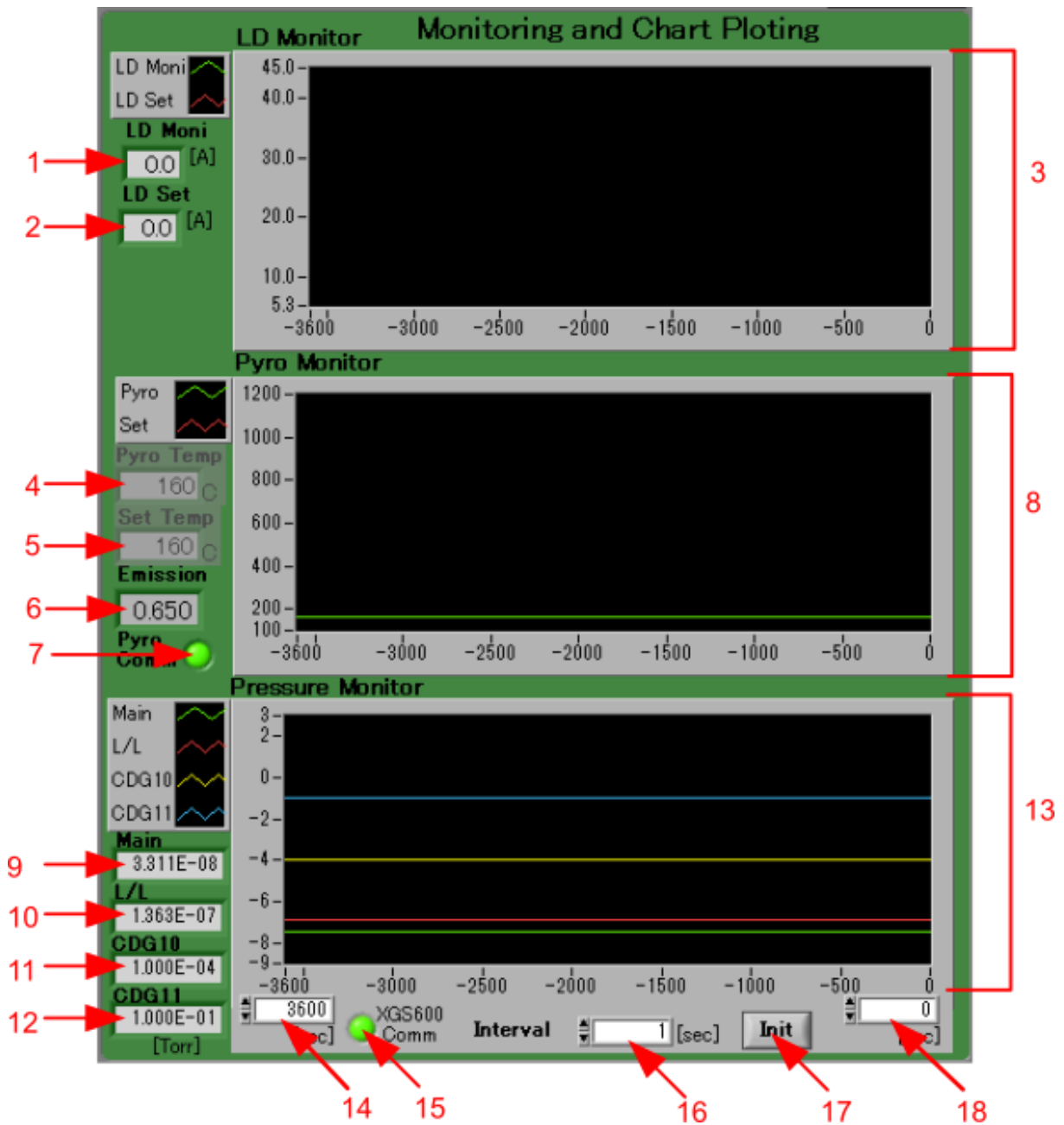


Figure 23:Status monitor panel.

Table 17: Functions in combinatorial mask positioning panel.

#	Names / Functions	Corresponding local operations
1	LD Moni display: the box shows present value (PV) of the laser driving current.	Local display is on front panel of laser-diode power supply.
2	LD Set display: the box shows set value (SV) of the laser driving current.	Local display is on front panel of laser-diode power supply.
3	LD time-chart: the graph shows temporary change in PV and SV of laser driving current.	No chart in local part.
4	Pyro display: the box shows present value (PV) of the substrate temperature measured by the pyrometer.	No display in local part.
5	Temp set display: the box shows set value (SV) of the substrate temperature defined at substrate heating control panel.	No display in local part.
6	Emissivity display: the box shows measurement parameter of pyrometer.	No indicator in local part.
7	Comm Status indicator: The indicator lights when signal/data communication is ready.	No display in local part.
8	Temperature time-chart: the graph shows temporary change in PV and SV of the substrate temperature.	No chart in local part.
9	Main-ch pressure display: it shows vacuum pressure in the main chamber.	The local indicator is on rack controller.
10	L/L-ch pressure display: the box shows vacuum pressure in the L/L chamber.	The local indicator is on rack controller.
11	Main-ch pressure display: it shows vacuum pressure in the main chamber by CMG10.	The local indicator is on rack controller.
12	Main-ch pressure display: it shows vacuum pressure in the main chamber by CMG11.	The local indicator is on rack controller.
13	Pressure time-chart: the graph shows temporary change in above pressure values.	No chart in local part.
14	Interval setting box: the box set the interval time of monitoring in unit of second.	No indicator in local operation part.
15	Comm Status indicator (Gauge): The indicator lights when signal/data communication is ready.	The local indicator is on rack controller.
16	Init button: the button clears monitor data and restarts.	No indicator in local operation part.
17	Plot size setting box: the box set full scale of the time chart in unit of second.	No indicator in local operation part.
18	Change size button: the button changes full scale of the time chart.	No indicator in local operation part.

The right bottom part in the main window is data logging part.

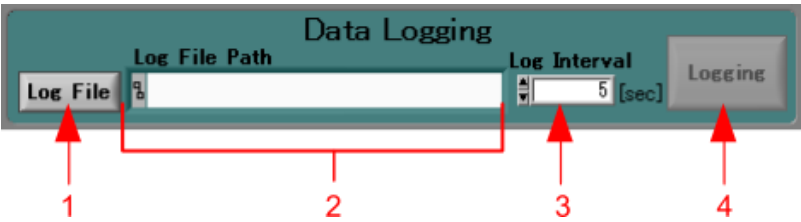


Figure 24:Data logging panel.

Table 18:Functions in combinatorial mask positioning panel.

#	Names / Functions	Corresponding local operations
1	Log file button: the button opens a sub-window to set a file name for saving log-data.	No indicator in local operation part.
2	Log file path setting box: the box set a path to access the log-file.	No indicator in local operation part.
3	Log interval setting box: the box set the interval time of data logging in unit of second.	No indicator in local operation part.
4	Start logging button: the button begins to write out the log data to a set log file.	No indicator in local operation part.

3.6 Supplemental operation

(1) Urgent/required motors' control and parameter settings on main window

The bottom part in the main window has supplemental operation for motorized motion of target manipulator, masks and sample. And the part set some parameters for the motorized motion and temperature control with PID setting.

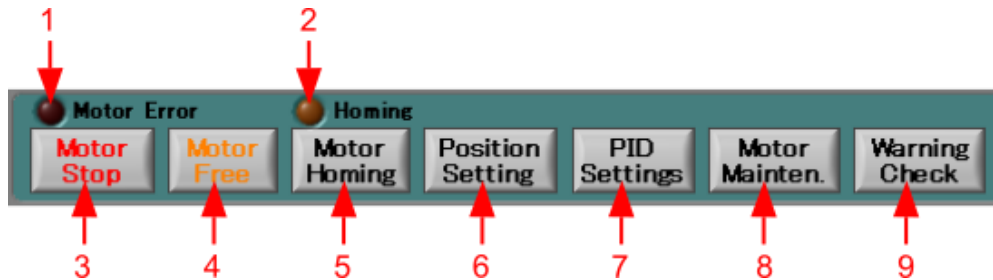


Figure 25: Supplemental control panel.

Table 19: Functions in supplemental control panel.

#	Names / Functions	Corresponding local operations
1	Motor Error indicator: the indicator lights when each motorized motion, the target manipulator, the masks, or the sample rotation, has trouble.	No indicator in local operation part.
2	Homing indicator: the indicator blinks when each motorized motion is in homing operation.	No indicator in local operation part.
3	Motor Stop button: the button is for emergency stop of motorized motion.	Local operation with handy console.
4	Motor Free button: the button sets out of motorized motion control without feeding power to motors.	Local operation with handy console.
5	Homing button: the button opens a sub-window for homing operation to a chosen motorized motion.	Local operation with handy console.
6	Position Setting button: the button opens a subwindow to set parameters for target position.	No local operation.
7	PID Setting button: the button opens a subwindow to set parameters for substrate heating.	No local operation.
8	Machine maintenance button: the button opens a sub-window to set parameters for motorized motion.	No local operation.
9	Warning check button: the button opens a subwindow to set critical point of warnings.	No local operation.

(2) Homing for motorized control objects

When pushing “Motor Homing” button, a sub-window opens for homing operation.

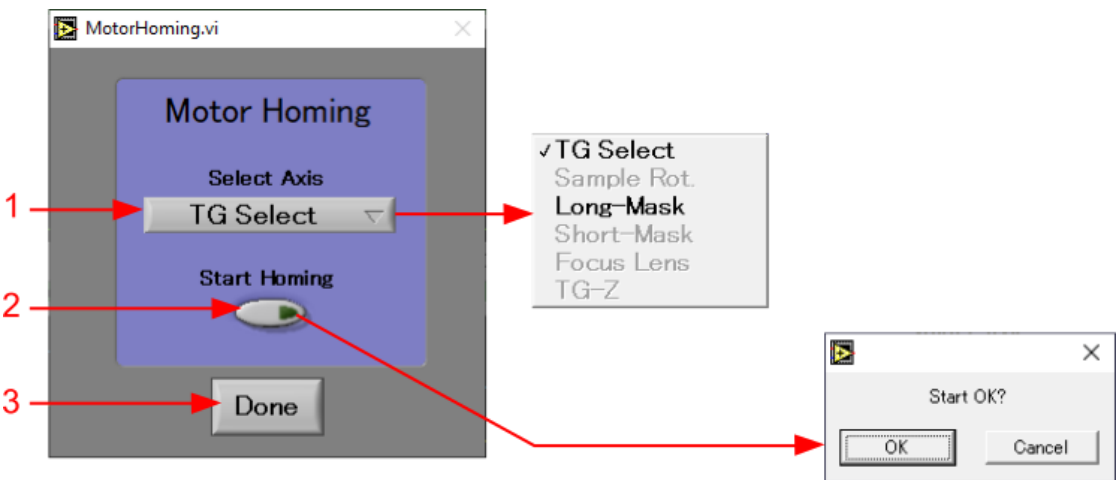


Figure 26:Homing operation sub-window.

Table 20:Functions in homing operation sub-window.

#	Names / Functions	Corresponding local operations
1	Motorized motion selector: the selector sets a motorized motion for homing by pull-down list.	Local operation with handy console.
2	Start button: the button starts homing operation.	Local operation with handy console.
3	Close button: the button closes the sub-window.	No local operation.

(3) Target position setting

When pushing “Position Setting” button, a sub-window opens for parameter setting.

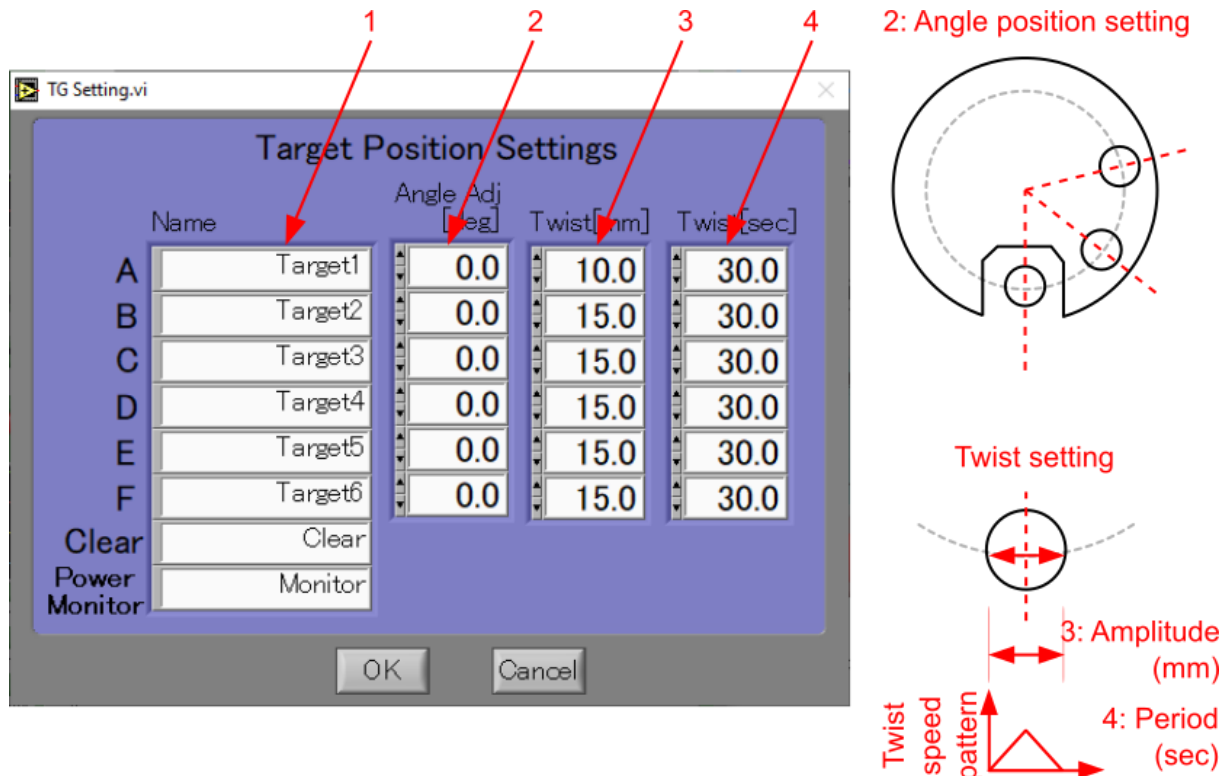


Figure 27: Target setting sub-window.

Table 21: Functions in target setting sub-window.

#	Names / Functions	Corresponding local operations
1	Target name setting-box: the box shows target names. The names can be modified.	No indicator in local operation part.
2	TG angle setting-box: the box sets angle positions of the targets. <i>Initial values are defined as factory setting.</i>	No indicator in local operation part.
3	Twist amplitude setting box: the box sets twist angle of the targets for uniform laser ablation. <i>Initial values are defined as factory setting.</i>	No indicator in local operation part.
4	Twist period setting box: the box sets twist period of the targets for uniform laser ablation. <i>Initial values are defined as factory setting.</i>	No indicator in local operation part.

(4) PID setting in sample heating

When pushing “PID Setting” button, a sub-window opens for parameter setting.

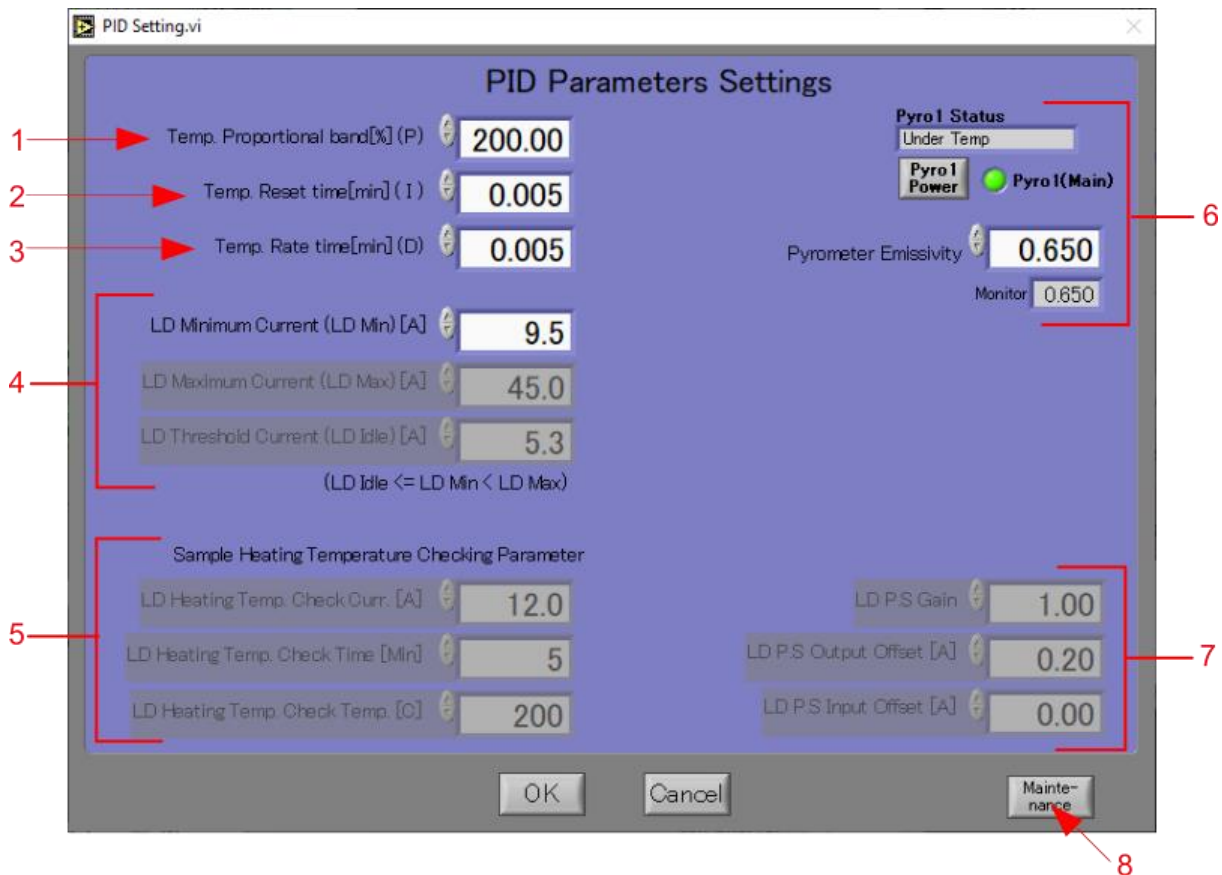


Figure 28:PID setting sub-window.

Table 22:Functions in target setting sub-window.

#	Names / Functions	Corresponding local operations
1	P-value setting box: the box sets proportional factor for feedback control of substrate temperature.	No indicator in local operation part.
2	I-value setting box: the box sets integral factor for feedback control of substrate temperature.	No indicator in local operation part.
3	D-value setting box: the box sets differential factor for feedback control of substrate temperature.	No indicator in local operation part.
4	LD current range setting boxes: the boxes set maximum, minimum and threshold driving-current of the laser diode with stable oscillation for stable PID temperature control.	No indicator in local operation part.
5	Temperature check parameter setting boxes: the boxes set critical point in laser-diode driving current and time response whether the pyrometer measure temperature properly.	No indicator in local operation part.

6	Pyrometer status monitor/setting boxes: the boxes show and set the pyrometer measurement parameter and working status.	No indicator in local operation part.
7	LD PS gain/offset setting boxes: the boxes set gain/offset of the laser-diode driving-current. <i>Initial values are defined as factory setting.</i>	No indicator in local operation part.
8	Maintenance button: the button enables to change factory settings in gray shading on the sub-window.	No local operation.

(5) Motor maintenance for motorized control objects

When pushing “Motor maintenance” button, a sub-window opens for parameter setting as shown in Figure 13.

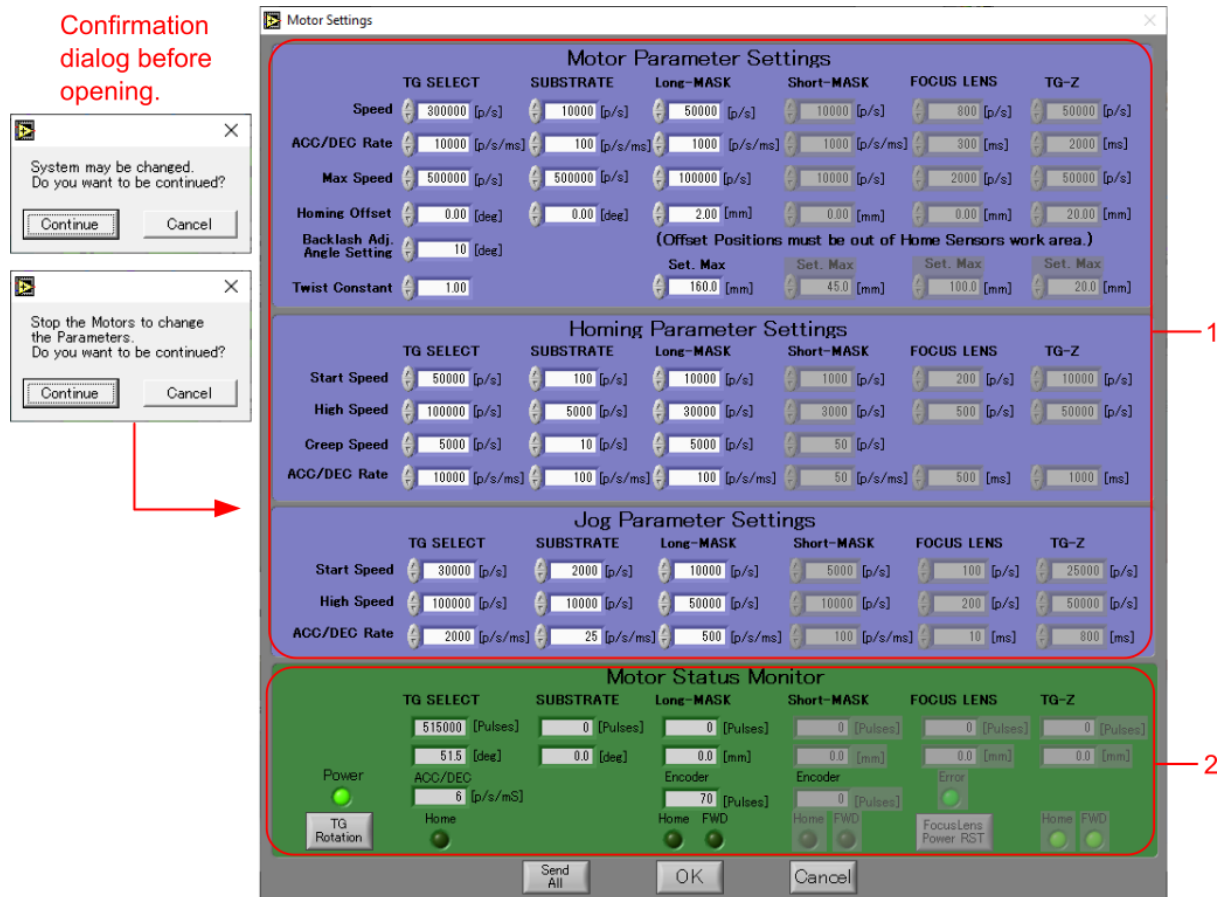


Figure 29:Motor maintenance sub-window.

Table 23:Functions in target setting sub-window.

#	Names / Functions	Corresponding local operations
1	Motorized-motion parameter-setting boxes: The boxes set motor speed for motorized motion of target manipulation or others during software operation, JOG control and homing. <i>Initial values are defined as factory setting.</i>	No indicator in local operation part.
2	Motor status monitor boxes: the boxes show current position value of motors/motorized motion factors. Unit of the motor position is in pulses and destination motorized position values are expressed in degree or millimeter.	No indicator in local operation part.

(6) Warning check

When pushing “Warning check” button, a sub-window opens for parameter setting.

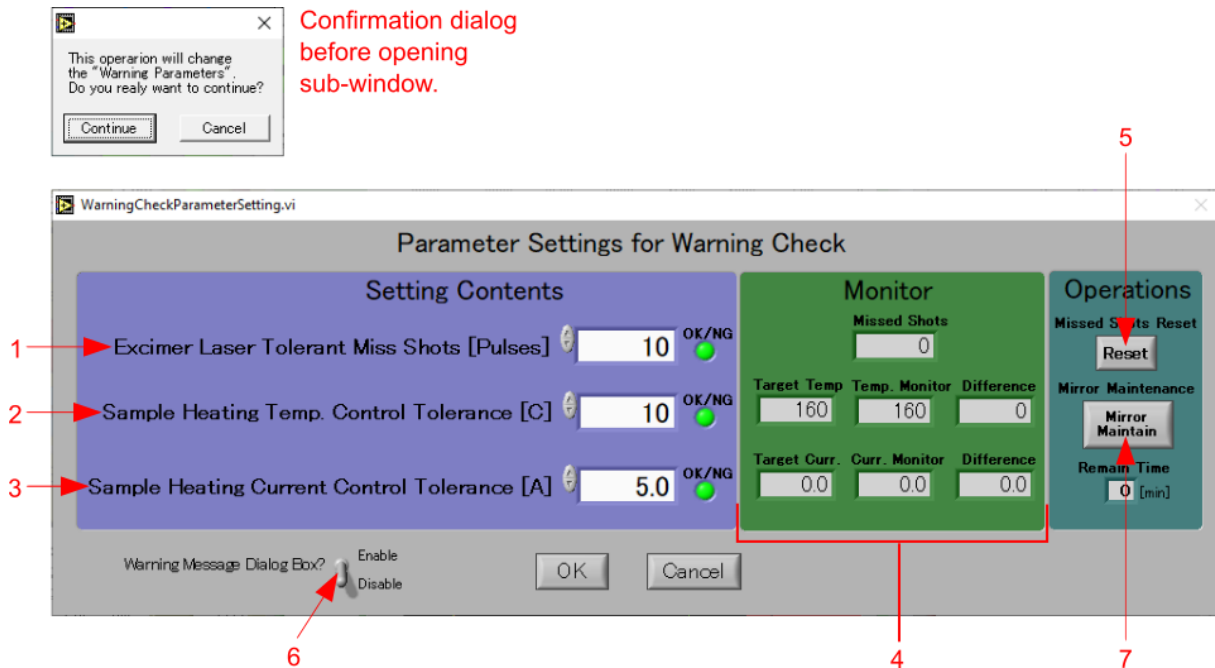


Figure 30:Warning check sub-window.

Table 24:Functions in target setting sub-window.

#	Names / Functions	Corresponding local operations
1	Excimer laser setting box: the box sets and shows permissible amount of miss-shots to total trigger signals during deposition.	No indicator in local operation part.
2	Temperature control setting box: the box sets tolerance to the set temperature value during PID temperature control.	No indicator in local operation part.
3	Current control setting box: the box sets tolerance of halogen-lamp current to the set value during manual control.	No indicator in local operation part.
4	Status monitor: these boxes show present values relevant to left critical values.	No indicator in local operation part.
5	Miss-shot reset button: this button set the ablation-laser miss-shot to zero.	
6	Warning message enabler: this switch sets whether the warning message dialog appears or not when above troubles and others.	
7	Mirror maintenance button: The button allows user to open the laser shield for alignment of laser beam path or others through two steps by entering password.	No indicator in local operation part.

System manual of PAC-LMBE

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